

Phase 7

Project Documentation Phase

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Phase	Covers
Brainstorming & Ideation Phase	Content management problem & proposed AI CMS solution
Requirement Analysis Phase	Software & hardware requirements
Project Design Phase	Backend layered architecture & AI service integration
Project Planning Phase	ER diagram, user flow, roles & MVC pattern
Project Development Phase	Backend setup, MongoDB connection, schemas & core code
Project Testing Phase	API endpoint testing & validation
Project Documentation Phase	Code structure, schemas & endpoints compilation
Project Demonstration Phase	Working video demo & live API execution

- **Problem & Requirements** – Explains the blog management and content creation challenges, the proposed AI BlogNest CMS solution, role-based usage, and required hardware/software.
- **Design & Architecture** – Describes the layered backend architecture, Entity-Relationship (ER) database structure, MVC pattern, user access roles (Admin, Editor, Author, Reader), and backend user flow.
- **Development & Implementation** – Covers VS Code project setup, Node.js/Express server configuration, MongoDB/Mongoose connection and schemas, JWT authentication, routing, and Gemini AI integration.
- **Testing & Demonstration** – Video demo covering backend setup and API testing for auth, blog management, and AI features.